

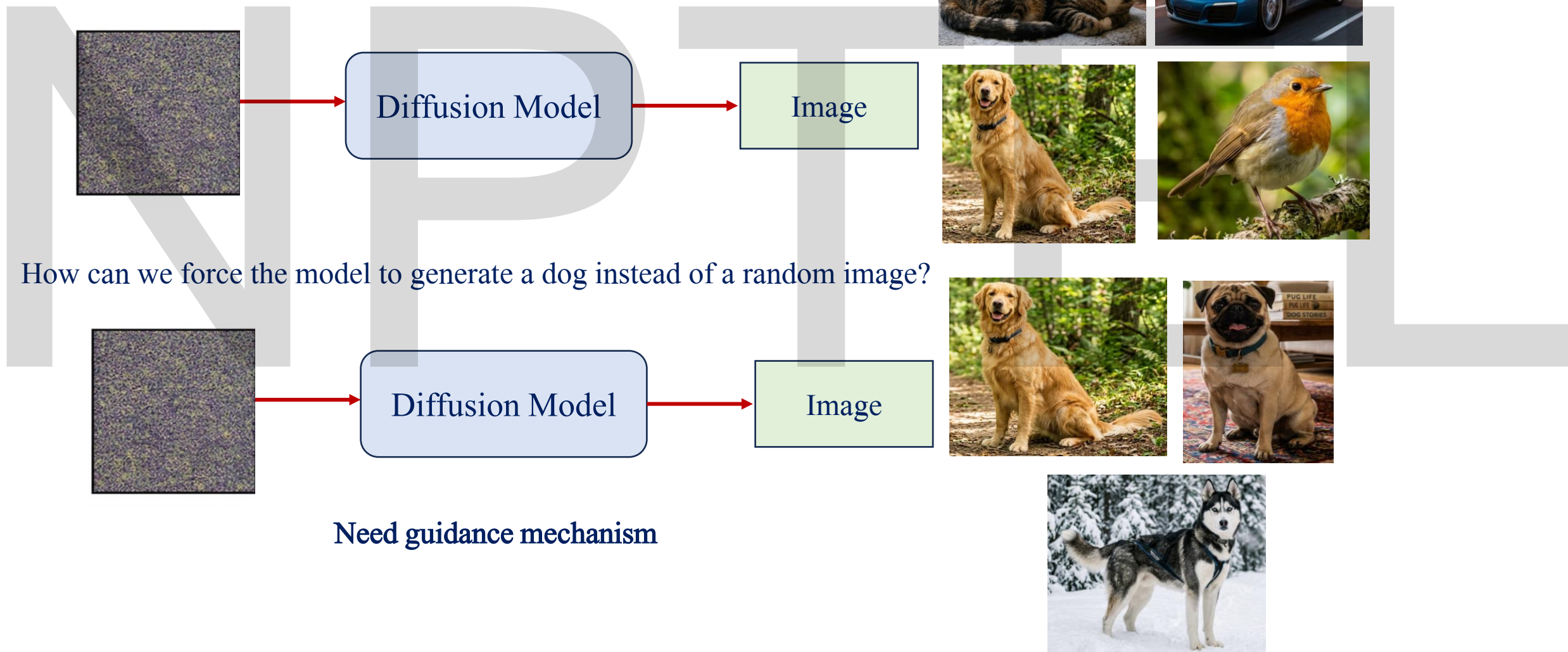
Classifier-Guided Diffusion Model

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Motivation

A standard DDPM learns the data distribution $p(x)$ and can generate realistic images, but it has no control over which class is generated.



Classifier-Guided Diffusion

Core idea

Take a generative model that knows how to create "anything" and give it a "GPS" to steer it toward a specific category.

Components of a Classifier Guided Diffusion

Diffusion Model



Acts as an artist that knows how to denoise an image into something realistic, but does not know which class to generate

Classifier



It acts as a critic which looks at the noisy image and predicts the probability of each class given a noisy image x_t

Classifier itself is a pretrained model.

Ref:

Dhariwal, P., & Nichol, A. (2021). Diffusion Models Beat GANs on Image Synthesis. NeurIPS 2021.

How it Works

Artist

Takes a small step to remove noise from the image

Adjustment

Update

Trade-off

Critic

Looks at that slightly clearer image and calculates the **gradient** of the class label (e.g., "dog") with respect to the image pixels.

✓ Classifier gradient indicates how to modify the image to increase the probability of a specific class (say y).

$$\nabla_{\mathbf{x}_t} \log p_{\phi}(y|\mathbf{x}_t)$$

The Diffusion model shifts its denoising direction toward that gradient.

Increasing the "guidance scale" makes the images much more recognizable as the target class (higher precision), but it often reduces the variety of images the model produces (lower diversity).

Orientation of Image Generation Towards a Particular Class

A DDPM learns the reverse transition:

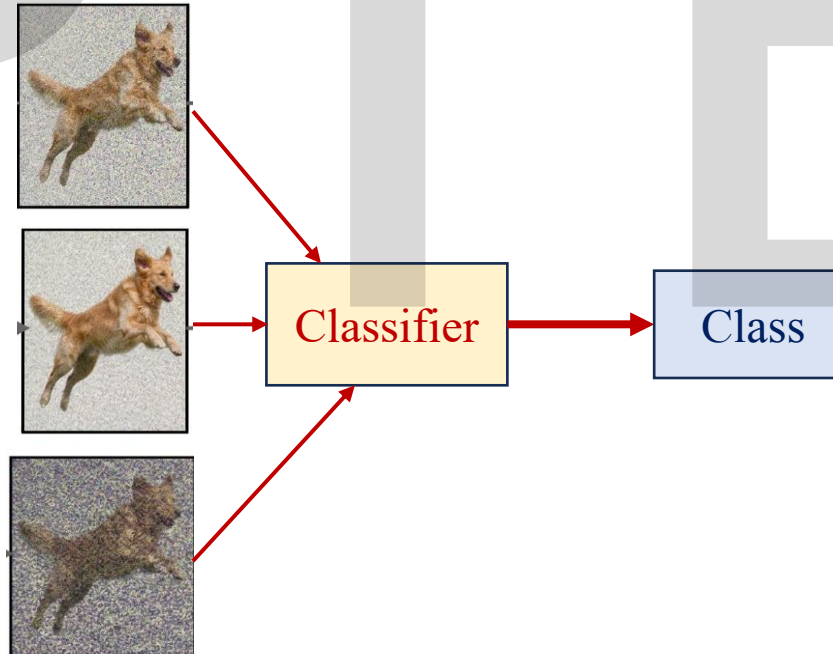
$$p_{\theta}(x_{t-1}|x_t) = \mathcal{N}(\mu_{\theta}(x_t, t), \Sigma_{\theta}(x_t, t))$$

Perturb the denoising trajectory in the reverse process to move it towards a particular class and then generate the image from that particular class.

Train a noisy image classifier

$$p_{\phi}(y | x_t)$$

Probability that noisy image x_t belongs to class y .

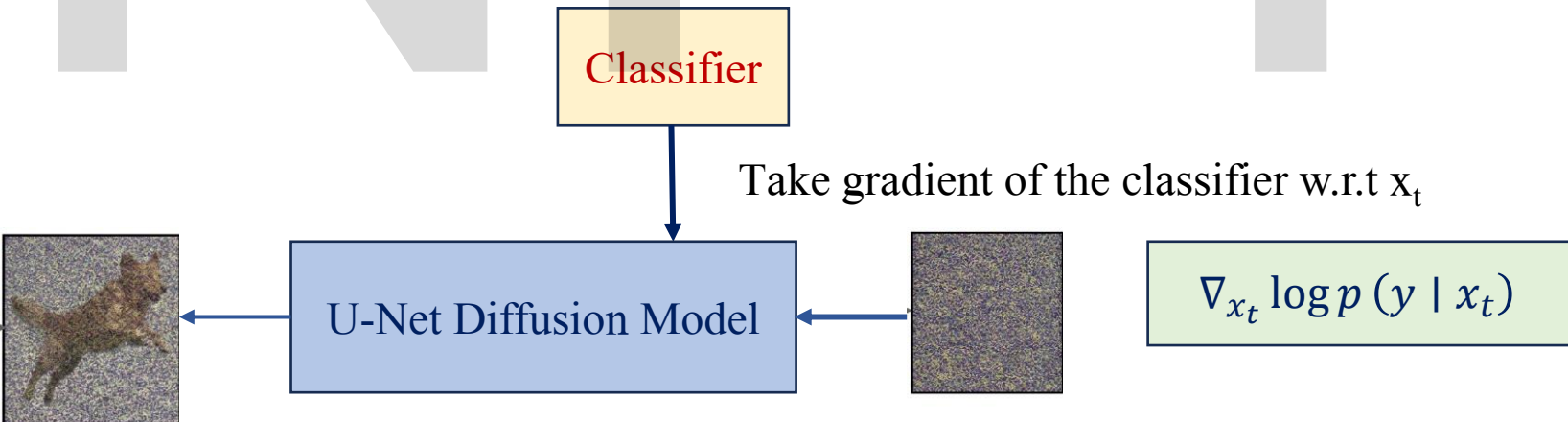


Sampling in Classifier-Guided Diffusion

Assume your DDPM is trained

During sampling:

- ✓ A noisy image is given to the network.
- ✓ The U-Net predicts the noise component $\epsilon_{\theta}(x_t, t)$ present in the noisy image.
- ✓ That amount of noise is removed from the image to get the clearer image.
- ✓ The gradient from the classifier is also given as input to the U-Net



Mathematical Formulation

We would like to sample from

$$p(x_t|y)$$

→ The probability of observing noisy image x_t conditioned on the class label y .

Using Bayes' theorem:

$$p(x_t | y) = \frac{p(y | x_t)p(x_t)}{p(y)}$$

Taking logarithm:

$$\log p(x_t | y) = \log p(x_t) + \log p(y | x_t) - \log p(y)$$

- ✓ Since $p(y)$ is constant (doesn't depend on x_t , its gradient w.r.t x_t is zero)
- ✓ Taking gradient with respect to x_t on both sides.

$$\nabla_{x_t} \log p(x_t | y) = \nabla_{x_t} \log p(x_t) + \nabla_{x_t} \log p(y | x_t)$$

This is the key equation of classifier guidance.

REALISTIC
IMAGE

TARGET
CLASS

$$\nabla_{x_t} \log p(x_t | y) = \nabla_{x_t} \log p(x_t) + \nabla_{x_t} \log p(y | x_t)$$

Gradient of the classifier

The unconditional DDPM already gives:

$$\nabla_{x_t} \log p(x_t) = -\frac{1}{\sqrt{1-\bar{\alpha}_t}} \epsilon_{\theta}(x_t, t)$$

Score = Constant $\times$ Noise Prediction

$$\bar{\alpha}_t = \prod_{i=1}^t \alpha_i$$

Represents the cumulative amount of original signal that remains after t forward diffusion steps.

$$\nabla_{x_t} \log p(x_t | y) = -\frac{1}{\sqrt{1-\bar{\alpha}_t}} \epsilon_{\theta}(x_t, t) + \nabla_{x_t} \log p_{\phi}(y | x_t)$$

Old Noise

New gradient term

Gives a new noise which is oriented towards a particular class

$$\nabla_{x_t} \log p(x_t | y) = \nabla_{x_t} \log p(x_t) + \nabla_{x_t} \log p(y | x_t)$$

$$-\frac{1}{\sqrt{1-\bar{\alpha}_t}} \hat{\epsilon}(x_t, t, y)$$

$$-\frac{1}{\sqrt{1-\bar{\alpha}_t}} \epsilon_{\theta}(x_t, t)$$

$$\hat{\epsilon}(x_t, t, y) = \epsilon_{\theta}(x_t, t) - s \sqrt{1-\bar{\alpha}_t} \nabla_{x_t} \log p_{\phi}(y | x_t)$$

s = guidance scale

The new noise is oriented towards a particular class

Classifier-guided noise prediction equation.

Small s : diverse but weak conditioning
 Large s : strong conditioning but less diversity

Sampling Equation

The guided noise estimate replaces the standard noise estimate in the DDPM reverse update

$$X_{t-1} = \sqrt{\alpha_{t-1}} \left(\frac{x_t - \sqrt{1 - \alpha_t} \hat{\epsilon}}{\sqrt{\alpha_t}} \right) + \sqrt{1 - \alpha_{t-1}} \hat{\epsilon}$$



s=1



s=100

Classifier guidance on the class “Pembroke Welsh corgi”. With classifier scale 1, results were not convincing, with scale =10.00 produces more class consistent images

Ref:

Prafulla Dhariwal ,Alex Nichol (OpenAI), Diffusion Models Beat GANs on Image Synthesis NeurIPS (2021)

Limitations of Classifier-Guided Diffusion

Requirement for a Separate Model:

- It requires training an additional classifier specifically on noisy images x_t to provide the necessary gradients at every step of the diffusion process
- Standard off the shelf classifiers like Resnet trained on ImageNet will fail as they are trained on clear images
- Complex pipeline as two separate networks (Unet/Diffusion network and the classifier) needs to be trained.
- Classifier gradients must be computed at every timestep, increasing computational cost during sampling

Adversarial Vulnerability

- Diffusion model looks for the easiest path to satisfy the classifier's gradient.
- This can lead to over-saturated images where the model generates artifacts that the classifier perceives as the target class, but don't look right to humans.

NPTEL

Summary

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Next Session

Classifier Free Guidance